

Almanac — Retrieval-Augmented Language Models for Clinical Medicine

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Abstract

BACKGROUND

Large language models (LLMs) have recently shown impressive zero-shot capabilities, whereby they can use auxiliary data, without the availability of task-specific training examples, to complete a variety of natural language tasks, such as summarization, dialogue generation, and question answering. However, despite many promising applications of LLMs in clinical medicine, adoption of these models has been limited by their tendency to generate incorrect and sometimes even harmful statements.

METHODS

We tasked a panel of eight board-certified clinicians and two health care practitioners with evaluating Almanac, an LLM framework augmented with retrieval capabilities from curated medical resources for medical guideline and treatment recommendations. The panel compared responses from Almanac and standard LLMs (ChatGPT-4, Bing, and Bard) versus a novel data set of 314 clinical questions spanning nine medical specialties.

RESULTS

Almanac showed a significant improvement in performance compared with the standard LLMs across axes of factuality, completeness, user preference, and adversarial safety.